ASUTOSH PATNAIK

IoT Engineer

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SUMMARY

A motivated engineering student with a strong foundation in electronics, digital circuits, and embedded systems, seeking an entry-level role in the electronics and technology industry. Possesses strong analytical and problem-solving skills, along with a keen eagerness to learn, contribute, and grow in a professional environment.

SKILLS

C C++ SQL Python

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

Centurion University of Technology and Management

2023 - 2027

CERTIFICATIONS

BIOTRIP 2026 – Certificate of Presentation – Agentic AI Orchestration

Presented a conference paper on *Agentic AI Orchestration* at BIOTRIP 2026, held at GIET University on February 25, 2026. Actively contributed to technical discussions on the role of agentic AI in medical research and received this certificate in recognition of the presentation.

PROJECTS

Home Automation System (2025–2026)

Developed a web-based home automation system using **WebSockets** for real-time bidirectional communication to remotely control smart door locks, automate lighting, and regulate fan speed based on temperature readings through a centralized dashboard.
Integrated **smoke, flame, and temperature sensors** along with a **buzzer** for fire safety, enabling continuous environmental monitoring and on-site audible alerts during emergencies.
Leveraged **WebSockets** for instant remote notifications and real-time dashboard updates, and **MySQL** for secure storage of user credentials, device status, access logs, and sensor event history, with robust authentication protocols ensuring data integrity.

Image Restoration using Restormer with ESP32-CAM

Built a web-based image restoration system utilizing the **Restormer architecture (CVPR 2022)** , a transformer-based model for high-quality deblurring, deraining, and denoising. Integrated **ESP32-CAM** for real-time image capture and wireless transmission, with a **Python** and **OpenCV** backend for processing degraded images. The web interface allows users to upload, restore, and view enhanced images, improving visual clarity for surveillance and low-quality imaging applications